The Use of Generative AI Tools in the Inception Stage of Software Startups

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Abstract. Generative Artificial Intelligence (GenAI) is disrupting numerous fields of human endeavors, including software startups. While GenAI tools hold the promise of accelerating innovation, improving product quality, and enhancing decision-making, software startups during their inception stage, a phase characterized by high uncertainty and limited resources, may face significant challenges in using these technologies. However, understanding the opportunities and challenges that GenAI brings to software startups is scarce due to the nascent nature of GenAI tools. Our study aims to provide an initial understanding of the intriguing research phenomenon of GenAI application in early-stage software startups. An action case study approach is employed in our study. We examined two software startups within a university setting, focusing on how these teams utilized GenAI tools in the inception stage. The findings revealed 11 opportunities and 10 challenges of using GenAI tools in the inception stage. Novel findings in the startup context include opportunities like brand identity development, generating landing pages, simulating customer feedback, and validating MVP. In contrast, challenges include a function-oriented rather than problem-oriented and a tendency to please rather than provide critical feedback. The findings suggest that while GenAI tools offer valuable benefits for startups, successful adoption requires careful consideration of technical and non-technical factors. This research opens avenues for future studies on integrating GenAI in early-stage software startups.

Keywords: Software Startup, Inception Stage, Generative AI, Action Case Study

1 Introduction

"Change the world for a better place" is a well-known motto that millions of startup companies worldwide embrace. Startups do not just adapt to the future — they create it. However, it is challenging to build a startup due to the inherent uncertainty associated with the process of "from 0 to 1" [16]. The failure rate is notoriously high [22]. Many startups fail to survive even the earliest inception

stage, from the initial conception of a business idea to the point where the startup delivers its first product to a customer [15]. This stage is crucial as the startup works toward achieving problem-solution fit, a milestone indicating that its solution adequately meets the needs of targeted customers [11].

The landscape of startups, including how they can be supported, is disrupted by the recent surge of Generative Artificial Intelligence (GenAI). According to a recent CBInsight report "Generative AI Bible: The ultimate guide to GenAI disruption"[3], an increasing number of startups pile into the GenAI arena, and funding soars as investors flock to catch the trend. In contrast, how GenAI is and will disrupt startups, especially in their early stages, receives much less attention, as the old saying depicts, "the darkest place is under the candlestick."

Like many other fields, GenAI technologies can enable startups to identify opportunities, accelerate innovation, and improve services and product quality, meeting the dynamic and challenging demands of the environments [20,19]. In the meantime, it is essential to address various issues and concerns associated with GenAI, such as bias in models and hallucinations [1], which could affect its applications in critical decision-making contexts. What opportunities and challenges GenAI technologies bring to startups and how startups can reap the benefits and handle the challenges are yet to be understood. These questions are particularly demanding for early-stage startups with limited resources and knowledge to experiment with GenAI, which prevents them from leveraging its potential to support their activities.

Our study aims to provide a good understanding of the potential of GenAI technologies in supporting startups. We focus on the inception stage of software startups. The inception-stage startups resemble infants in their vulnerability and need for support to thrive. They rely heavily on various resources such as incubation programs, mentoring services, and networking opportunities [6]. However, these resources are often limited and not universally accessible, leaving many inception-stage startups without the tailored support they require to address their unique challenges and increase their survival rates [18]. The rationale behind the focus on software startups is that they typically adopt cutting-edge technologies early and are highly influenced by them [10].

In this study, we ask the following research question:

RQ: How is the usage of GenAI technologies perceived by software startups in the inception stage?

To answer the research question, we applied the action case approach [4] and studied two software startups in the university settings in depth. The findings from our study include eleven opportunities and ten challenges associated with the key steps in the inception stage of software startups. The two cases provided a contextual understanding of these opportunities and challenges.

The rest of the paper is organized as follows. Section 2 reviews background literature and existing research relevant to our study. The use of action case as the research methodology is explained in Section 3, and the data collection and analysis processes are detailed in the same section. The findings are reported in

[3] https://www.cbinsights.com/research/report/generative-ai-bible

Section 4 and discussed in Section 5. Section 6 concludes the paper with future work.

2 Literature Review

2.1 Inception Stage of Software Startups

Software startups are the startups in which software plays a central role in the innovations they are working on [17]. Limited resources, limited experiences, and the strong influence of cutting-edge technologies are some of the key characteristics of software startups [24].

A recent study by Klotins et al. [15] models the software startup lifecycle into four stages: inception, stabilization, growth, and maturity. In the inception stage, software startups focus on discovering gaps and developing innovative ideas into viable solutions [15]. Bosch et al. [3] model early-stage software startup development into three phases: idea generation, ideas backlog prioritization, and funneling through problem validation, solution validation, and minimum viable product (MVP) validation on small and large scales.

Studies show that developing viable solutions during the early stages of a software startup is a challenging process [10,25]. For example, Giardino et al. [10] explore the most common challenges early-stage software startups face during the initial period. These challenges include technological issues, customer acquisition, funding, team development, defining MVP, and product-market fit.

2.2 GenAI for Innovation

GenAI refers to AI technologies capable of creating new content [8], such as text, images, audio, video, 3D, and code. These technologies use advanced machine learning techniques and neural networks to analyze vast information and generate original outputs that mimic human-generated content [8]. In the software industry, GenAI has been used to streamline software development processes through task automation, such as image generation, code generation, testing, debugging, deployment, expert replacement, and documentation [12,13,23]. Conversely, this technology has various challenges, including bias, hallucination, and inaccurate responses [12,13,23].

Studies point out the potential use of GenAI to foster and accelerate innovations [2,9]. GenAI can help companies in the initial stage of their innovation initiatives, from ideation to the solution validation phase [9]. In the ideation phase, GenAI can assist idea generation from nothing or from an analysis of existing data, e.g., customer reviews and behavior, to capture patterns and uncover problems from which new ideas emerge. Later, GenAI can also assist innovation teams in evaluating these new ideas and creating assumptions, i.e., hypotheses. Following that, GenAI can help with product validation purposes, which benefits teams without tech-savvy personnel [9]. Moreover, GenAI also has the potential to assist in organizational, managerial, and customer-related activities such as customer support and analytics for recommendation systems [9].

Apart from these few studies that explore the potential of GenAI in the software industry and to support innovation in companies, to the best of our knowledge, there is no study on the opportunities and challenges brought by GenAI to software startups. Our study intends to bridge this knowledge gap to advance research and practice in the context of software startups.

3 Research Approach

To answer the research question, we applied the action case research method, a hybrid of interpretation and intervention, understanding and change [4]. Action case lies between case study and action research. It introduces small-scale intervention with a deep contextual understanding of a case. We deem action case as an appropriate research method for this study because, when studying a nascent research phenomenon such as GenAI, which is not widespread yet, a researcher needs to take a more active role and be involved in creating change in practice rather than being an observer only.

This study involved two software startups in a university setting. Both teams worked on the startup projects for four months during the data collection in June 2024. The two software startups were in the inception stage since neither has reached a problem-solution fit yet.

One of the authors intervened with the two startup teams by encouraging the teams to utilize GenAI technology to support their startup development. Several tools were recommended, including ChatGPT, the GPT builder (to create digital personas representing early adopters), and mixo.io (automated landing page generation from a stated idea). The teams were not obliged to use the recommended tools and had the liberty to use any GenAI tools they considered appropriate. Additionally, the teams were mentored by university lecturers and local entrepreneurs.

3.1 Data Collection

The primary data collection method was semi-structured interviews with open-ended questions. Since one of the authors was familiar with the two software startup teams already, we kept the part related to the demographic and background of the interviewees minimal. The main part of the interview was guided by the steps that represent the typical steps a startup follows in the inception stage, according to the Lean Startup methodology [21] and the literature reviewed in Section 2.1. The first step, **ideation and problem identification**, combines idea generation with ideas backlog prioritization, emphasizing problem identification. The second step is **problem validation**, intending to understand whether the hypothesized customer problems do exist. The third step, **problem-solution fit**, focuses on validating the proposed solution.

For each of the steps, we asked the following open-ended questions in the interviews:

- (Q1) Which GenAI tools did your team use?

– (Q2) How did your team use them?
– (Q3) What are your opinions about the tools used?

Additional questions emerged during the interviews and were followed up wherever possible. The interview protocol used can be accessed online[4].

All team members were interviewed, and their profiles are summarized in Table 1. The table provides background information on each interviewee. Members from the Rapido team are indicated with the prefix **R**, while those from the Blue Dot team are marked with **B**. Each interviewee within their respective teams is assigned a unique identifier in the format **I*n*** (where n represents the interviewee's sequence number).

Table 1. The Profiles of Interviewees

Team	**Interviewee**	**Background Information**
Rapido	R-I1	R-I1 is an experienced engineer in front-end, back-end, and full-stack roles. R-I1 has interned in a company based in Italy, working with TypeScript, and holds a bachelor's degree in computer science. R-I1 has experience developing mobile applications and has used Figma for mock-ups.
	R-I2	R-I2 is a software developer with experience in both front-end and back-end development from university projects. R-I2 specializes in back-end development and works with C#. Moreover, R-I2 has also developed mobile applications. R-I2 holds a bachelor degree in computer science.
	R-I3	R-I3 have experience working as an engineer for websites and mobile applications, with occasional front-end tasks. R-I3 has intervened in a company with a role related to compliance assessment.
Blue Dot	B-I1	B-I1 is a full-stack developer, proficient in both back-end and front-end development. B-I1 uses Scala for back-end tasks and PureScript for front-end development.
	B-I2	B-I2 has experience with Java, HTML, Python, Node.js, and jQuery for website development and JavaScript for app development. B-I2 has also learned PHP through an internship and has experience writing smart contracts using Solidity, which is based on C++.

Three of the four authors were present in all the interviews conducted. The author who intervened with the two teams by introducing GenAI tools to them was the main interviewer. The other two authors acted as observers and asked

[4] Interview protocol: https://doi.org/10.5281/zenodo.14036029

additional questions when appropriate. All interviews were conducted online using Microsoft Teams[5]. The duration of the interviews varies between 30 minutes to 45 minutes per interviewee. All interviews were transcribed using OpenAI's Whisper[6]. The transcriptions were reviewed, and missed transcribed texts were manually fixed based on the recordings.

In addition to the interviews and observations during them, documents produced by the two teams were collected, including pitch decks, promotion videos, recorded pitches, landing pages, filled lean canvas, learning diaries, and custom GPTs. These documents and observations were used to triangulate the data collected through the interviews.

3.2 Data Analysis

We conducted a thematic analysis of the interview data following the approach outlined by Braun and Clarke [5]. We also followed the guidelines provided by Cruzes et al. [7] to synthesize our findings. Our analysis was both deductive and inductive. Initially, we developed a set of high-level codes aligned with the research question. The first and second authors took primary responsibility for analyzing the interview transcripts. Both authors then independently read the transcripts to understand the content and context. The two authors reconciled their coding, and all authors reviewed the final results for rigor and consistency. The documents, unlike interview data, were analyzed to gain insights into GenAI tool usages.

4 Findings

This section describes how the two startup teams used various GenAI tools and how they perceived the opportunities and challenges associated with the usage of GenAI. Table 2 presents the GenAI tools utilized by the two startups.

Table 2. Reported GenAI tools used by the two startups

Name	**Type**	**Usage in the Cases**
Gamma.app	Text-to-Image	Create presentations for pitch decks
logomaster.ai	Text-to-Image	Design logos for products or brands
UIZard.io	Text-to-Image	Develop app mockups for user testing and feedback
ChatGPT.com	Text-to-Text	Validate ideas, generate brand names, custom GPTs, and content
Wix.com	Text-to-Text	Build and customize websites for companies
FlexClip.com	Text-to-Video	Create video presentations for pitching ideas
InVideo.io	Text-to-Video	Create video presentations for pitching ideas

The findings are structured using the three steps described in Section 3.2. Table 3 overviews the opportunities and challenges identified in the two cases.

[5] Microsoft Teams: https://www.microsoft.com/en-us/microsoft-teams
[6] OpenAI's Whisper: https://openai.com/index/whisper

Table 3: Opportunities and Challenges of using GenAI Tools

		Case 1. Rapido			**Case 2. Blue Dot**		
		I&PI	**PV**	**P-S Fit**	**I&PI**	**PV**	**P-S Fit**
Opportunities	Op01. Brand identity development	✓					
	Op02. Create mockups and websites			✓			✓
	Op03. Create presentation			✓			
	Op04. Develop interview questions		✓				
	Op05. Domain expert replacement			✓			✓
	Op06. Generate image, audio, and video			✓			
	Op07. Generate landing page			✓			
	Op08. Idea generation and refinement				✓		
	Op09. Language translation		✓				
	Op10. Simulate customer feedback			✓			
	Op11. Validate MVP			✓			✓
Challenges	Ch01. Biased responses		✓				
	Ch02. Cannot represent the actual users		✓			✓	
	Ch03. Felt unnatural			✓			✓
	Ch04. Pricy		✓			✓	
	Ch05. Hard to construct meaningful prompts		✓				
	Ch06. Inaccurate responses		✓				
	Ch07. Function-oriented instead of problem-oriented		✓				
	Ch08. Tend to please instead of being critical					✓	
	Ch09. Lacks of transparency						✓
	Ch10. Unreliable responses		✓				

Note: I&PI: Ideation and Problem Identification; PV: Problem Validation; P-S Fit: Problem-Solution Fit
The highlighted cells indicate that the corresponding challenges are not specific to any steps.

4.1 Opportunities and Challenges Perceived by Rapido

The Rapido team (Table 1: R-I1, R-I2, and R-I3) proposed a digital platform to revolutionize the booking process for sports facilities such as tennis courts and football pitches. This innovative solution replaces traditional manual, paper-based methods with a streamlined and efficient workflow. By integrating online booking capabilities, the platform offers users a convenient and user-friendly experience, allowing them to reserve sports fields effortlessly.

Ideation and problem identification In this step, the team brainstormed and developed **brand identity (Op01)** by using ChatGPT, R-I1 shared, *"We were trying to come up with a few names based on the idea's description. We went through some options, but in the end, we picked the one we liked best"* and created some logos as R-I1 shared, *"It's an AI tool for creating free logos, and there are a lot of websites offering something similar"* from which the team then chose one of the given alternatives as their product logo.

Problem validation In this step, the team used GenAI to **develop customer interview questions (Op04)**. In this use case scenario, they asked ChatGPT to suggest relevant questions mentioning *"essentially, we wanted to figure out which questions would give us the most useful information. Since the goal of these interviews was validation, we needed to focus on questions that would provide insights we found valuable"* as R-I1 stated. Then using these questions, they interviewed several sports field owners and customers about their experiences and challenges while managing and making reservations.

Some customer interviews were conducted with potential users who speak Spanish due to the connection one team member had back in his home country Peru. Since not all team members are Spanish speakers, they used AI to translate these interview transcriptions into English. In this early step, GenAI's **language translation (Op09)** capability proved useful for the team.

Despite the opportunities, the custom GPTs also brought challenges. Firstly, they potentially generate **biased responses (Ch01)** *"because they were shaped for our needs,"* R-I3 stated. Secondly, the team felt that the custom GPTs **cannot fully represent the actual users (Ch02)**, making validation necessary as R-I3 suggested to *"go into the real world, and it may be different because every person has a different opinion."*

Problem/solution fit In this step, the team used GenAI to help them develop their solution, starting from **creating presentations (Op03), images, audio, video (Op06), mockups (Op02), and landing page (Op07)** using AI-generated prompts. The power of GenAI helped the team fill the need for **expertise in particular fields (Op05)**, in this case, while developing the landing page.

The team utilized two custom GPTs to gather more detailed human-like opinions about their MVPs, allowing them to seek immediate **customer feedback**

through simulation (Op10) before collecting input from actual users. This approach was particularly useful as finding actual was challenging, and they *"did not intend to go to ask actual people"* as R-I3 mentioned for a halfway solution, making the simulated interaction a practical alternative. In this case, the team used the custom GPTs to **validate their MVP (Op11)**, such as their landing page and video. The custom GPTs developed by this group can be found in Appendix A.

Despite the opportunities, GenAI's outputs were sometimes **felt unnatural (Ch03)**, R-I3 shared an example of this challenge, *"the generated images sometimes were kind of unnatural, and we can see that they are AI-generated."*

Challenges across all steps in the inception stage The first challenge was **high price (Ch04)**, R-I3 stated *"these [features] are more advanced and you have to pay for them. Since we do not have much money. It's difficult"* for the team to enjoy the full potential of GenAI tools and had to be satisfied with limited customizations and watermarks. Unfortunately, it was **hard to construct meaningful prompts (Ch05)**, R-I3 stated, *"prompting the right term to the right input is crucial ... We need to know very well what to be written is exactly what is in our mind."* Otherwise, **inaccurate responses (Ch06)**, R-I3 continued, *"you can get some results that are not so aligned with your mind, your idea."* Another reported challenge was the available GenAI tools were more **function-oriented instead of problem-oriented (Ch07)**. R-I1 argued that instead of having separate tools for image, logo, and text generation to develop a webpage, *"it would be great if we could find a tool that takes all of them in one place."* The team also experienced GenAI produced **unreliable responses (Ch10)**, *"sometimes the response is good, sometimes it needs small adjustments, and sometimes it is unacceptable,"* R-I3 shared.

4.2 Opportunities and Challenges Perceived by Blue Dot

The Blue Dot team (Table 1: B-I1, and B-I2) was building a community-based travel platform to facilitate ride-sharing for short distances within urban areas. This solution addresses several common urban transportation challenges, including difficulties in finding public transport in certain areas and the scarcity of parking spots for private vehicles. By enabling people to share rides, the platform aims to enhance urban mobility, reduce traffic congestion, and promote a more sustainable mode of transportation.

Ideation and problem identification In this step, the team used GenAI tools for **idea generation and refinement (Op08)**. They found ChatGPT particularly effective at generating ideas and refining concepts, noting that it was *"very good at generating stuff and coming up with ideas,"* as B-I1 mentioned. This approach was especially valuable in the inception stage of a startup, providing crucial guidance that helped them shape their initial ideas.

Problem validation In the problem validation step, the team developed a custom GPT and considered the possibility of validating the hypothesized customer problems using the custom GPT. However, the team pointed out that they were worried that the custom GPT **cannot represent the actual users (Ch02)**, emphasizing the need to *"go and talk to [real] people to understand the problem,"* B-I1 shared, highlighting the limitation of GenAI in replacing direct user interaction for problem validation purpose. The custom GPTs developed by this group can be found in Appendix A.

Problem/solution fit In this step, the team used GenAI tools like UIZard[7] to **create mockups (Op02)** as B-I1 mentioned *"We briefly used AI to create the app mockup because the software we were using only accepts prompts. For example, we enter something like build a ride-hailing app,"* they generated basic screens that helped visualize the application.

The team also identified an opportunity for **domain expert replacement (Op05)**, where B-I1 mentioned *"If you are developing a software solution, GenAI can be a big help in creating an initial prototype. If you do not know how HTML, CSS, or JavaScript work, especially for those without a technical background,"* also offering assistance to **validate MVP (Op11)** by refining the content for their product as mentioned by B-I1 *"On Netlify, I believe we selected a template. In the beginning, we may have used the persona to help validate some of the text being added to our MVP website"*.

Despite this, the team experienced challenges when creating MVPs, such as promotional videos and landing pages. The responses generated by GenAI **felt unnatural (Ch03)**, with videos appearing *"strange and a bit unsteady"* as mentioned by B-I2. Furthermore, there were **lacks of transparency (Ch09)** in GenAI tools, where the team perceived it as a *"black box magic machine that brings out code"* as mentioned by B-I1.

Challenges across all steps in the inception stage One of these challenges was GenAI is **pricy (Ch04)** as premium subscriptions were required for premium features. This was not affordable for the team, and they had to create multiple trial accounts to enjoy premium features. B-I1 shared an example of this challenge, *"I think I had to create four different accounts because we changed ideas. As a student trying to save money."* Additionally, GenAI tools tended to **please the users instead of being critical (Ch08)**, B-I1 shared, *"I think AI tends to please and agree with you."*

4.3 Cross-case Comparison

The use of GenAI tools provided various opportunities to the two studied software startups, particularly in areas such as creating mockups and validating

[7] UIZard: https://uizard.io

MVPs. The ability to quickly generate mockups (Op02) was perceived as beneficial by both teams, enabling them to rapidly visualize their ideas without needing extensive design expertise or resources. This efficiency saved time and allowed the teams to iterate on their designs more effectively. Similarly, using GenAI to validate MVPs (Op11) offered a streamlined approach to gathering feedback and making necessary adjustments before presenting the product to actual users. This capability was particularly valuable for early-stage startups, where resources are limited, and the need for quick, cost-effective validation processes is critical.

However, these opportunities were also accompanied by several common challenges. Both teams encountered issues with the limitations of GenAI, such as the inability to fully represent actual users (Ch02), leading to concerns about the accuracy and relevance of the feedback generated by GenAI tools. Additionally, the output generated by GenAI sometimes felt unnatural (Ch03), which affected the overall quality of the prototypes and MVPs. The high cost of premium GenAI tools (Ch04) was another shared challenge, particularly for startups operating on tight budgets. These challenges underscore the need for careful consideration when integrating GenAI into the early stages of product development, balancing the benefits of efficiency and innovation with the potential drawbacks of cost and accuracy.

There was a noticeable difference in the confidence levels regarding using GenAI tools. **Rapido** team demonstrated a strong confidence in the potential of GenAI tools, integrating these tools extensively into their workflow, trusting the technology to enhance their startup development. Their confidence in GenAI led them to explore and leverage GenAI tools capabilities. In contrast, **Blue Dot** team was more hesitant and relied heavily on their own experience, expressing significant doubts about the effectiveness of GenAI tools. Their skepticism made them cautious in adopting GenAI, and they were more comfortable relying on their expertise. This team is concerned about the risks and limitations GenAI tools might pose, which affected their willingness to integrate them into their startup development.

5 Discussion

The inception stage of software startups is marked by high uncertainty and limited resources, making it a critical period for innovation and strategic decision-making. To address the research question, **How is the usage of GenAI technologies perceived by software startups in the inception stage?**, findings from the action case study of two inception-stage software startups reveal that GenAI tools can offer substantial **opportunities** in this phase by providing functionalities that streamline processes, enhance creativity, and accelerate product development, while also introduces significant **challenges** that must be carefully managed.

The eleven opportunities and ten challenges identified in this study can be viewed from both business and technical perspectives. From the business per-

spective, opportunities involve evaluating how GenAI can address the startup's core business problems, while the technical perspective focuses on the functionalities that GenAI offers. For example, Team **Rapido** used GenAI to create a landing page (Op07), a prototype of their solution (i.e., business aspect), without having to write codes manually (i.e., technical aspect).

These perspectives are natural, given that the primary goal of a software startup is to develop viable business ideas while managing the technical aspects of implementation. Consequently, the identified opportunities can be categorized into business opportunities (Op01, Op02, Op03, Op04, Op05, Op07, Op08, Op10, Op11) and technical opportunities (Op06, Op09). Likewise, the challenges fall into business challenges (Ch02, Ch04, Ch07, Ch08) and technical challenges (Ch01, Ch03, Ch05, Ch06, Ch09, Ch10). These opportunities and challenges are highly relevant to the startup context, aligning with the critical role of GenAI in driving business model innovation, which is particularly crucial for startups striving to maintain competitiveness [14].

From our analysis, we found that several opportunities (Op02, Op03, Op04, Op05, Op06, Op08, Op09) and challenges (Ch01, Ch03, Ch04, Ch05, Ch06, Ch09, Ch10) have also been reported in prior studies [12,13,23], thereby consolidating the existing knowledge related to GenAI. However, we also identified novel opportunities such as brand identity development (Op01), generating landing pages (Op07), simulating customer feedback (Op10), and validating MVP (Op11). Additionally, we discovered novel challenges where existing GenAI tools are more function-oriented than problem-oriented (Ch07) and tend to please rather than provide critical feedback (Ch08). These novel findings are particularly pertinent to the startup context, offering new insights into how GenAI can be both a powerful tool and a potential pitfall in the early stages of software startup development.

Limitation A potential threat to the internal validity of our findings is the intervening nature of the action case study. One of the authors introduced and encouraged the teams to use GenAI tools in their startup development. This may have influenced the teams to be more positive towards the use of GenAI tools to support their startup activities. To mitigate this potential threat, we avoided any leading questions and tried our best to keep a neutral tone during the interviews. We also triangulated interviewees' opinions using the documents and observations we gathered as part of the action case study.

As with any case study, a limitation of this research is the generalizability of the findings. A small number of cases were studied. The opportunities and challenges identified from the two studied cases may be incomplete and not representative for the general population of software startups. The study's exploratory nature and the multiple-case design mitigated this validity threat to a certain extent.

Another limitation of the study is that it focused on inception-stage software startups within a specific university setting, which may not reflect the experiences and challenges of software startups in industries, even though it is not

uncommon for startup teams to initiate their startup development journey while they are university students. Further investigation is necessary to explore the use of GenAI tools across a broader range of startup ecosystems. This would help understand the full potential and limitations of GenAI tools in various contexts, providing a more comprehensive view.

6 Conclusion

GenAI became a disruptive force to all walks of life only less than two years ago. We are yet to fully grasp its potential to support startup endeavors. Our study is one of the first attempts that aspires to bring insights on how to utilize GenAI effectively in the startup context. It highlighted the use of GenAI tools in the inception stage of software startups, presenting significant opportunities and challenges. While GenAI can accelerate innovation, streamline product development, and offer cost-effective solutions, its application has risks, such as generating biased outputs, high costs for premium features, and difficulties in fully representing actual user needs. The findings underscore the need for early-stage startups to strategically integrate GenAI, balancing its potential benefits with carefully considering its limitations.

Following this initial step, we will study more early-stage software startups to consolidate and extend the findings reported in this study and map specific GenAI tools to relevant opportunities and challenges through multiple case studies and survey studies. We also plan to cover other stages of software startups, including stabilization, growth, and mature stages, in our future research.

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Appendix

A Custom GPTs

The Rapido team created two custom GPTs backed with the two identified personas and used these custom GPTs to validate the problem further. The team derived two personas based on the customer interviews. One is called Laura Fajitas, representing the sports field owners. Laura "*is 33 years old, Peruvian-Italian, manages football field reservations in Pescara, Italy*". The other is called Alex Sportivo, representing a sports facility user. Alex is *"a 27-year-old young man in Italy who supports Modena FC and Real Madrid, and whose favorite tennis player is Rafael Nadal. Alex plays as a left-footed defender in soccer and enjoys tennis to unwind"*.

Similarly, the Blue Dot team created a persona called Luca to represent potential platform users and built a custom GPT to represent Luca digitally. Luca is "*a student and part-time worker living in Ravenna. He juggles his university classes in the morning and work in the afternoon, all while trying to balance life with his girlfriend. He commutes around the city a lot and tries to avoid using his car whenever he can because finding parking is a pain, and fuel is expensive. When he's not working or studying, he either hangs out with friends or runs errands like grocery shopping and cooking dinner at home*".